<u>Change Report</u>

ENG1 Cohort 1 Team 3

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Change Management, Tools, Processes and Convention

This section will describe the methods, tools, and approaches the team uses to manage and track the Assessment 1 deliverables and documentation inherited from the previous team's project. To ensure optimal results, every change made to the code and documentation will be reported and tracked systematically. To plan the project, the team will hold weekly meetings to discuss task assignments, ensuring that each member is clear on their responsibilities and deadlines. This approach promotes collaboration and accountability throughout the process.

Documentation

Google Docs was selected for document editing and updates because of its collaborative features and ability to track version history. Each document is assigned a specific reviewer, responsible for providing feedback and suggesting improvements on parts of the content that might require more explanation or expansion. For diagrams, PlantUML was used as it can be integrated well with GitHub, our chosen version control system, as well as allowing dynamic diagram generation.

Code

We used GitHub as our main version control system to handle changes to the codebase. The decision was based on GitHub’s intuitive branching and commit features, which allowed the team to work on different aspects of the project at the same time without interfering with each other. Any updates such as new methods, classes, or other alterations to the inherited code, were successfully documented and commented within the code itself. This process helped us keep everything ensured and transparent that everyone was on the same page regarding the adjustments made. Also, through GitHub Actions, we managed to integrate CI pipelines with our codebase.

Requirements

Original Document:	Updated Document:
Req1.pdf	Req2.pdf

After reviewing the system, we identified key areas for improvement. Our main priorities are to make the game design clearer and usable, while ensuring it aligns better with what users expect and what is technically achievable.

Change 1:

Two new user requirements were added:

- UR_ACHIEVEMENTS: "The game shall include unlockable achievements to track player milestones."
- UR_SCORES: "The game shall maintain a score system reflecting user's satisfaction."

Justification:

In order to improve user engagement and to provide a sense of accomplishment particularly in a positive way, these user requirements were added.

Impact:

The game mechanics will change as the user is capable of tracking their satisfaction and achievements. This requires additional design, implementation and testing efforts but provides a better experience..

Change 2:

Three new functional requirements were added:

- FR_AUDIO: "The game shall include background music and sound effects, preferably adjustable through settings."
- FR_ACHIEVEMENTS: "The game shall track achievements, which are milestones reached by players during gameplay."
- FR_SCORES: "The game shall maintain a score reflecting user's satisfaction based on buildings, gameplay events and choices."

Justification:

On top of UR_ACHIEVEMENTS and UR_SCORES, these will enable the system to improve user engagement, as well as enabling the system to provide audio feedback for a more immersive experience.

Impact:

Increases the scope of development and also adds an audio aspect to the program.

Change 3:

One new non-functional requirement was added:

- NFR_RESPONSIVENESS: "The system should be able to resize on different laptop screens"

Justification:

To accommodate a wider audience and a broader market, the game should be compatible with many (particularly standard but not limited to) screen sizes.

Impact:

This increases the required amount of testing and also expands the implementation phase to provide responsiveness.

Architecture

Original Document:	Updated Document:
Arch1.pdf	Arch2.pdf

Description

Main objective of the update in architecture is to introduce the changes made in the design and to reflect the structure with more elaborate diagrams. Along with this, we also introduced new scenarios that are newly implemented in accordance with the updated requirements.

Change 1:

Introduction of Events, Achievements and CooldownTimer.

Justification:

To reflect the expanded scope of the requirements, we implemented additional requirements' necessities.

Impact:

Increases the code and architectural complexity, but provides better game experience.

Change 2:

Improved class diagrams, state diagrams and sequence diagrams for scenarios.

Justification:

New components and features were introduced that require being shown by the class architecture more articulately. These features also changed the flow of the game that needed to be reflected in the state diagram and those features resulted in the indication of new scenarios.

Impact:

The architecture is now easier to understand just by the class diagram. The flow of the states are updated and the new scenarios are explained clearly.

Method Selection and Planning

Original Document:	Updated Document:
Plan1.pdf	Plan2.pdf

Description

Engineering is always about the trade-offs and when it comes to picking a methodology or a tool, there are always alternatives. Those methodologies and tools were compared to help to justify our choices. Following discussions of how and with what to work, we focused on role assignment and planning.

Change 1:

Changed the communication medium from Discord to WhatsApp.

Justification:

As our team is more used to WhatsApp, we chose it over Discord to eliminate possible communication mistakes and/or possible outcomes of not being able to communicate quickly.

Impact:

Reduced communication overhead and kept the whole team up-to-date.

Change 2:

Assigned roles and responsibilities to individuals

Justification:

To be able to have the separation of responsibilities, we needed to define clear roles and assign them to individuals.

Impact:

Improved role alignment with requirements while adding some communication complexity.

Change 3:

Weekly plannings were completed beforehand

Justification:

Created gantt charts for weekly plannings in accordance with individuals' responsibilities. Also created breakdowns of the issues.

Change 4:

Continuous Integration

Justification:

To reduce the issues related to code integration and to ensure the quality of the code GitHub Actions were used.

Impact:

Helped better collaboration between individuals and assured the quality of the code, yet added an extra layer of technical setup.

Risk Assessment and Mitigation

Original Document:	Updated Document:
Risk1.pdf	Risk2.pdf

Changes to the risk report were made to provide clearer explanations and stronger justification for the risk management strategies used throughout the project. Additionally, new risks were added to ensure all potential challenges were addressed and to provide a more comprehensive view of the project's risk landscape.

Change 1:

Introduced new types to the Risk Register

Justification:

The previous version didn't take software, hardware, human, and estimation issues into account. Therefore, we needed to expand the scope of the possible risk sources. The new types might not be as likely as the former ones, yet they would help us mitigate broader types of risks. Such as, some graphics might be triggering for some cultures and user testing would prevent such a case from becoming widespread. Or, some graphics or some technical parts of the game might be IP protected, which requires you to be more careful about the originality of the code and the graphics or their licences.

Change 2:

Introduced new mitigation strategies

Justification:

As new types of risks were discovered, those inherently required new ways of mitigations or avoidance. Such as, compatibility issues can be avoided by testing on different platforms and/or different devices. Or unexpected crashes can be mitigated by aggregating error logs, using automated tests, conducting through code reviews and continuous feedback collection.

Change 3:

The risk register was updated to reflect the changes in team members, with the "Owner" field adjusted accordingly to ensure accurate assignment of responsibilities.

Justification:

As we take on the project from the other team, it's essential to reassign the ownership of the risks. This will help ensure that they are closely monitored and effectively managed as we move forward.